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DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H2 BIOLOGY

9648/01

Paper 1 Multiple Choice Questions

23 September 2013

1 hour 15 mins

Additional Materials: OTAS sheet

INSTRUCTIONS TO CANDIDATES:

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

Section A MCQ [40 marks]

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

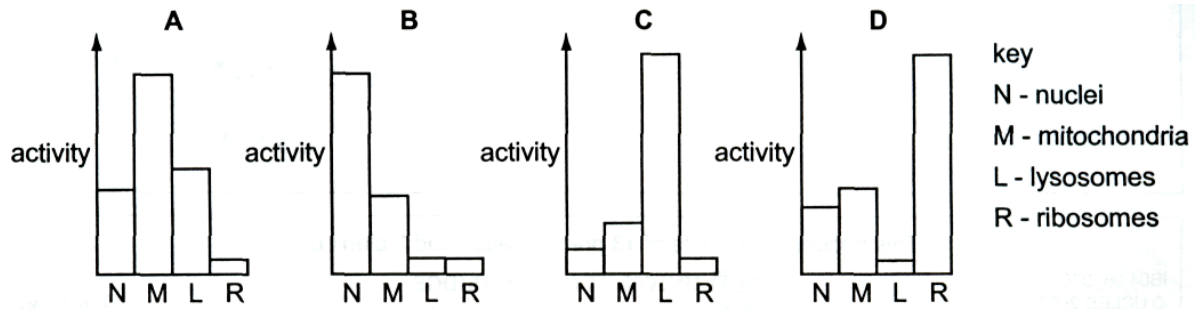
Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

This document consists of **20** printed pages and **0** blank page.

Answer **all** questions in this section.

- 1 A piece of mammalian tissue was homogenised and subjected to differential centrifugation to yield four subcellular fractions. The activity within each fraction, of four different types of enzyme, **A**, **B**, **C** and **D**, was investigated.

Which bar chart shows the results of investigating hydrolytic enzyme activity?



- 2 Membranes in cells include the following components.

- 1 cholesterol
- 2 glycoproteins
- 3 phospholipids
- 4 proteins

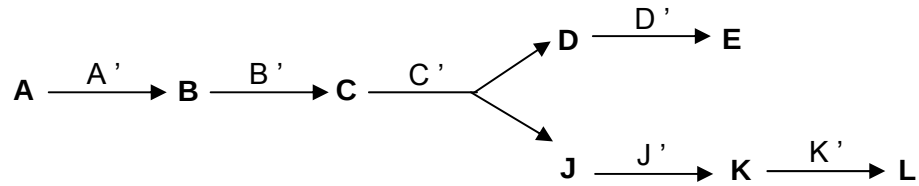
Which component is the most important for these functions of membranes?

function	recognising self/ non self	separating dissolved ions	stabilising the hydrophobic layer	transporting ions through membranes
A	1	2	3	4
B	2	3	1	4
C	3	1	4	2
D	3	4	1	2

- 3 In amylopectin, the bonds found between glucose units are _____.

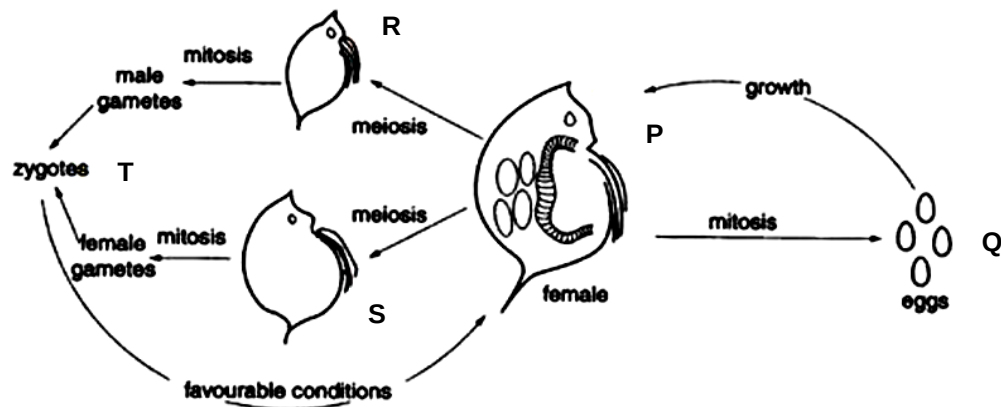
- A** α (1 $\rightarrow$ 4) glycosidic bonds only
- B** α (1 $\rightarrow$ 4) glycosidic bonds and α (1 $\rightarrow$ 6) glycosidic bonds only
- C** β (1 $\rightarrow$ 4) glycosidic bonds only
- D** β (1 $\rightarrow$ 4) glycosidic bonds and β (1 $\rightarrow$ 6) glycosidic bonds only

- 4 The figure shows a series of reactions in a metabolic pathway.



Enzyme C' catalyses the splitting of C into D and J . Assuming that product E is an allosteric inhibitor of C' , which of the following would likely happen if E were not consumed in a subsequent reaction?

- A The rate of production of D would increase.
 B The rate of production of E would remain the same.
 C The rate of production of L would remain the same.
 D The rate of production of all products D , E , J , K would decrease.
- 5 The figure shows the life cycle of the water flea, *Daphnia*. The cells of individual R contain 10 chromosomes.



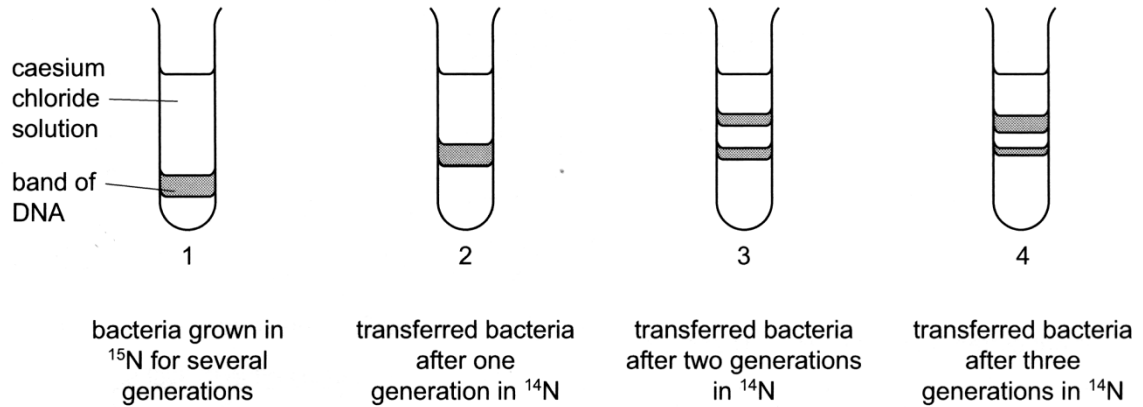
Which of the following are correct?

	Individual	Ploidy level	Number of chromosomes	Reason for choice
I	P	$2n$	20	The cells of P can undergo both mitosis and meiosis.
II	Q	$2n$	20	P produces eggs Q by mitosis which develop into females.
III	S	n	10	The gametic cells of P have undergone normal meiosis.
IV	T	$2n$	20	Random fertilisation of haploid gametes from R and S occurred to form zygote T .

- A I and II only
 B I and III only
 C II and IV only
 D All of the above

- 6 Bacteria grown in ^{15}N for many generations were transferred to ^{14}N for further replication. DNA from the bacteria was extracted and separated by density gradient centrifugation.

Their results are summarised in the following diagram.



What are the percentages of DNA containing ^{15}N and ^{14}N in tube 4?

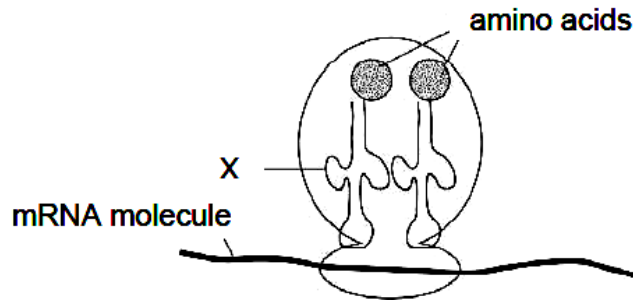
	percentage DNA	
	^{15}N	^{14}N
A	0	100
B	10	90
C	12.5	87.5
D	15	85

- 7 What is the basis for the difference in the synthesis of the leading and lagging strands of DNA molecules?

- I The anti-parallel arrangement of the DNA strands.
- II The RNA primers are required to initiate DNA elongation.
- III DNA polymerase joins new nucleotides to the 3' end of the growing strand.
- IV Helicase and single-stranded binding proteins work at the 5' end of the DNA strand.

- A** I and III
- B** II and IV
- C** I and IV
- D** II and III

- 8 The diagram below shows part of a molecule of mRNA bound to a ribosome.



Which of the following is **FALSE** about molecule X?

- I. It is formed by RNA polymerase in the nucleus.
- II. It is able to form hydrogen bonds with mRNA.
- III. An amino acid was attached to it by the enzyme amino-acyl tRNA transferase.
- IV. It is held in the amino-acyl tRNA binding site of the ribosome.

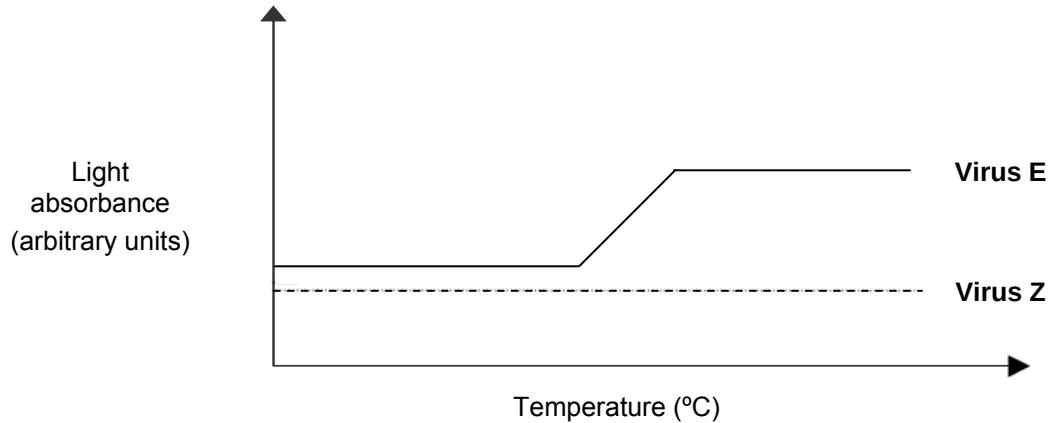
- A III only
- B IV only
- C I and II
- D III and IV

- 9 Which of the following shows the possible effects of a single nucleotide substitution in each of the following locations in a gene on the production of the protein it codes for?

	Promoter	Transcription terminator	Start codon	Stop codon	Middle of an intron
A	No protein product is produced	Protein product is shorter than normal	Protein product is longer than normal	Protein product is normal	Too much protein product is produced
B	Too much protein product is produced	Protein product is normal	No protein product is produced	Protein product is longer than normal	Protein product is normal
C	Protein product is normal	Protein product is longer than normal	Protein product is shorter than normal	Too much protein product is produced	Protein product is longer than normal
D	Protein product is longer than normal	Too much protein product is produced	Protein product is normal	Protein product is shorter than normal	No protein product is produced

10 Two new **viruses E** and **Z**, which infect eukaryotic cells, have been identified.

In one experiment, the nucleic acid from each virus is isolated and analyzed over a range of temperatures. The light absorbance of nucleic acids changes when denaturation or annealing occurs. The behaviour of the nucleic acid from each virus is shown in the graph.

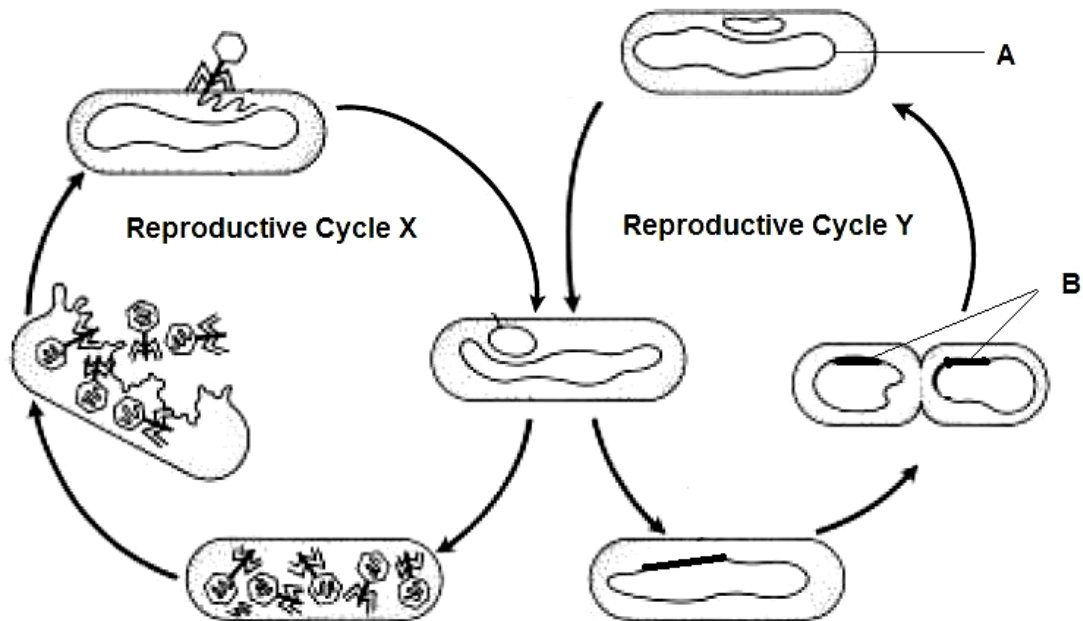


In a second experiment, it is found that treatment with reverse transcriptase inhibitors or with inhibitors of DNA synthesis blocks the ability of **virus Z** to infect cells. In contrast, reverse transcriptase inhibitors have no effect on the ability of **virus E** to infect cells but DNA synthesis inhibitors block infection by **virus E**.

Which of the following conclusions can be drawn from both the experiments?

- A** The genome of **virus E** is single-stranded RNA and that of **virus Z** is double-stranded DNA.
- B** The genome of **virus E** is double-stranded DNA and that of **virus Z** is single-stranded RNA.
- C** The genome of **virus E** is double-stranded RNA and that of **virus Z** is single-stranded DNA.
- D** The genome of **virus E** is double-stranded DNA and that of **virus Z** is double-stranded RNA.

- 11 Which of the following correctly identifies the reproductive cycles X and Y and structures A and B?



	X	Y	A	B
A	lytic	lysogenic	virus DNA	provirus
B	lysogenic	lytic	virus DNA	prophage
C	lytic	lysogenic	bacterial DNA	prophage
D	lysogenic	lytic	bacterial DNA	provirus

- 12 Bacterial strain A has been infected with viruses. Upon lysis, the virions produced are introduced to Bacterial strain B. After a short period of time, a new strain of bacteria is detected that is very similar to strain A but has a few characteristics of Strain B. Which is the process that leads to the production of the new strain?

- A Conjugation
- B Transduction
- C Transformation
- D Transposition

- 13** A student has two cultures of cells. Cell culture **A** consists of a culture of normal cells whereas cell culture **B** consists of a culture of cells dividing uncontrollably (isolated from a tumour). He wanted to determine if the uncontrolled growth (in culture **B**) was the result of an oncogene or a mutated tumour suppressor gene. He decided to fuse the cells from the two cultures (i.e. culture **A** with culture **B**) to form a hybrid cell line. As he was not certain of the expected results, he wrote down all the possible combinations in terms of the expected results and the gene that is involved as shown below.

	Expected results of hybrid cells	Gene involved
1	Grows normally	Oncogene
2	Grows uncontrollably	Oncogene
3	Grows normally	Mutated tumour suppressor gene
4	Grows uncontrollably	Mutated tumour suppressor gene

Of the four possibilities, only two of them are set correctly. Based on your understanding of the genetic changes that has occurred in oncogenes and the mutated tumour suppressor genes, choose one of the options below which shows the two correct possibilities.

- A** 1 and 3
B 1 and 4
C 2 and 3
D 2 and 4

- 14** Four different genes are regulated in different ways.

- Gene 1 undergoes tissue-specific patterns of alternative splicing.
 Gene 2 is part of a group of structural genes controlled by the same regulatory sequences.
 Gene 3 under some circumstances undergo methylation.
 Gene 4 codes for a repressor protein which acts at an operator site close by.

Which row of the table correctly identifies which genes are prokaryotic and which are eukaryotic?

	prokaryotic	eukaryotic
A	1 and 2	3 and 4
B	1 and 3	2 and 4
C	2 and 3	1 and 4
D	2 and 4	1 and 3

15 Which of the following contains repetitive DNA sequences?

- I. Telomere
- II. Centromere
- III. Intron
- IV. Control element

- A** I and II
- B** II and III
- C** III and IV
- D** I, II and III

16 A cross between a round-leafed, tall plant and round-leafed dwarf plant produced the following offspring:

121 round-leafed, tall plant
 124 round-leafed, dwarf plant
 42 oval-leafed, tall plant
 37 oval-leafed, dwarf plant

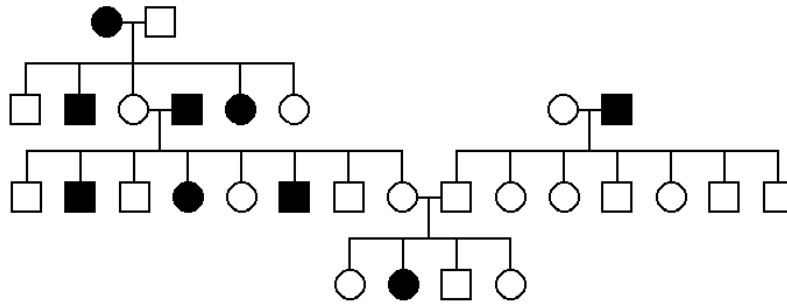
Key

R – round leaf
r – oval leaf
T – tall
t – dwarf

What were the genotypes of the parents?

- A** $RrTt \times Rrtt$
- B** $RrTt \times RRtt$
- C** $RrTT \times Rrtt$
- D** $RrTT \times RRtt$

- 17 The inheritance of a disease in a family is presented in a pedigree tree below.



What type of inheritance could this disease show?

- I Autosomal dominant
 - II Autosomal recessive
 - III Sex-linked dominant
 - IV Sex-linked recessive
- A I only
- B II only
- C I and III
- D II and IV
- 18 Agoutic mice have banded hairs, giving a grey colour. Black mice have unbanded hairs. White mice have no pigment. A cross between a homozygous black mouse and a white mouse produced offspring with agouti hair. Another cross between the black mouse and another white mouse produced some offspring with agoutic hair and some with black hair.

What explains these observations on the phenotype of hair of mice?

- A There is a single gene with two codominant allele, black and white.
- B There is a single gene with three alleles in a dominance series, black $\rightarrow$ grey $\rightarrow$ white.
- C There are two epistatic genes, one controlling pigment production and one controlling banding.
- D There are two linked genes, one controlling pigment production and one controlling banding.

- 19**
1. Light joins carbon dioxide to an acceptor compound which is then reduced by hydrogen obtained from water.
 2. Carbon dioxide combines with an acceptor compound and this is reduced by hydrogen split from water by light
 3. Light splits carbon dioxide and the resulting carbon then combines with oxygen and hydrogen obtained from water.
 4. Light splits water and the resulting hydroxyl group combines with a compound which has incorporated carbon dioxide

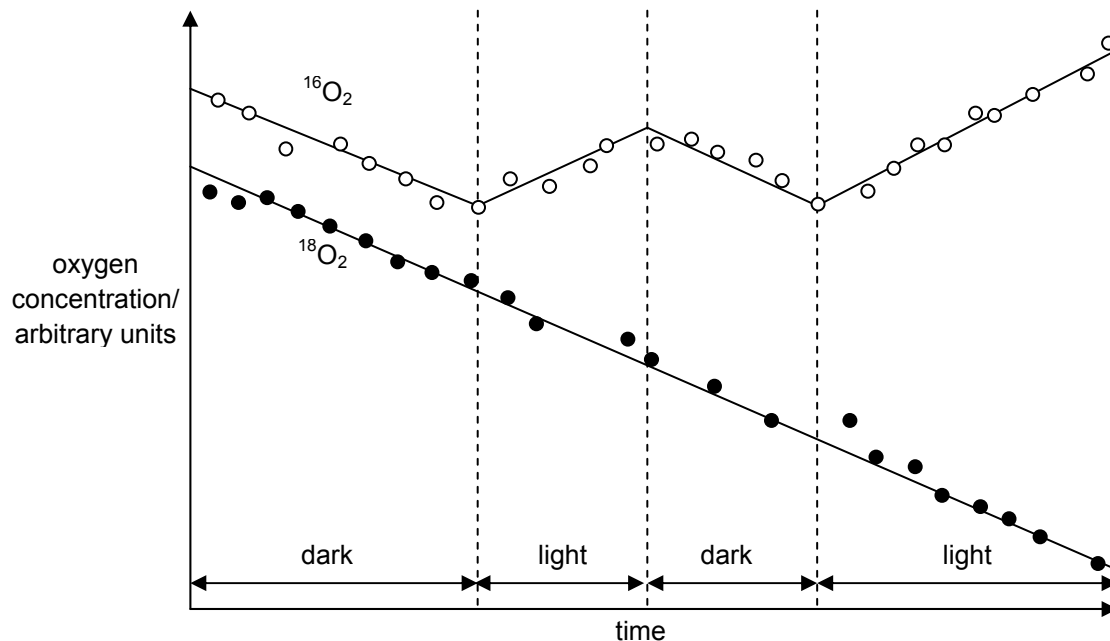
Which of the following statement is/are **CORRECT** outline of the main events in photosynthesis?

- A** 1 only
- B** 2 only
- C** 1 and 2
- D** 3 and 4
- 20** Dinitrophenol is a metabolic poison that can lodge within the thylakoid membranes of chloroplasts. It then provides an alternative route for H^+ ions to diffuse across the thylakoid membranes. In what way will the Calvin cycle be affected in chloroplasts poisoned by dinitrophenol?
- A** No change in rate as Calvin cycle occurs in the stroma and not at thylakoid membranes.
- B** The rate of Calvin cycle will increase as pH in the stroma will decrease towards the optimum for enzymes involved in the cycle.
- C** The rate of Calvin cycle will decrease with the accumulation of glycerate-3-phosphate.
- D** The rate of Calvin cycle will decrease with the accumulation of glyceraldehyde-3-phosphate.

- 21 Isotopes of oxygen can be used to distinguish between oxygen absorbed by plants and oxygen evolved.

A mixture of oxygen isotopes $^{16}\text{O}_2$ and $^{18}\text{O}_2$ was supplied to a suspension of the unicellular alga *Chlorella* which had previously been exposed to $^{16}\text{O}_2$ only. During the following hour, changes in concentration of these gases in the suspension were measured in light and dark conditions.

The graph shows the results.



What caused the concentration of $^{16}\text{O}_2$ to rise in light?

- A H_2^{18}O was being photolysed more rapidly than H_2^{16}O .
 - B $^{16}\text{O}_2$ was being absorbed at different rates in light and dark.
 - C $^{16}\text{O}_2$ was being produced in photosynthesis but was not being absorbed in respiration.
 - D $^{16}\text{O}_2$ was being produced in photosynthesis faster than it was being absorbed in respiration.
- 22 Both glucose and appropriate enzymes are necessary for the process of glycolysis to begin.

Which additional compound must also be present?

- A Acetylcoenzyme A
- B ATP
- C Pyruvate
- D Reduced NAD

23 In which of the following setups would carbon dioxide be produced?

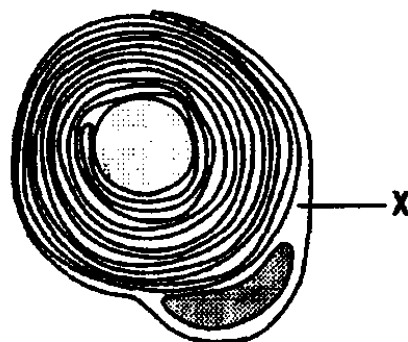
	I	II	III
A	Mitochondria + glucose	Cell lysate + pyruvate	Chloroplast + RuBP carboxylase
B	Mitochondria + starch + amylase	Cell lysate + RuBP carboxylase	Cell lysate + pyruvate
C	Cell lysate + glucose	Cell lysate + pyruvate	Mitochondria + pyruvate
D	Cell lysate + pyruvate	Chloroplast + RuBP carboxylase	Lysosome + starch

24 Cyanide is a poison that prevents the production of ATP.

What will happen if a resting axon is treated with such a poison?

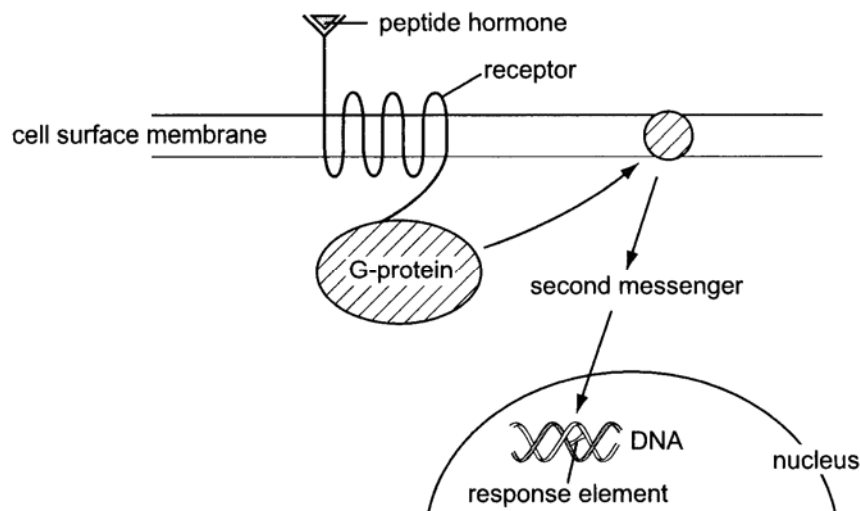
- A** The charge outside the axon membrane does not change.
- B** The charge inside the axon membrane becomes more negative.
- C** The charge outside the axon membrane becomes less positive.
- D** The charge inside the axon membrane becomes less positive.

25 The diagram represents a transverse section through a neurone as seen under an electron microscope. Which of the following most appropriately describes the function of **X**?



- A** Insulate the neurone to maintain homeostatic conditions.
- B** Provide mechanical protection to the axon.
- C** Prevents ion exchange between the axon and extracellular fluid.
- D** Speeds up nervous transmission by allowing impulse to travel along it.

- 26 A particular type of autoimmune disease causes antibodies to destroy acetylcholine receptors of neurons. What effect will this have on the functioning of the nervous system?
- A Presynaptic neurons will be unable to release neurotransmitters into the synaptic clefts.
 - B Neurons will be unable to propagate action potentials along their axons.
 - C Depolarized neurons will be unable to reestablish an ionic gradient across their membranes.
 - D Postsynaptic neurons will be unable to detect signals from presynaptic neurons.
- 27 These statements describe an action potential. Which one of the following is **FALSE**?
- A Its speed can be increased by myelination of the neurone.
 - B Its depolarization phase is an example of a positive feedback.
 - C It is regarded as an all-or-nothing response.
 - D It is initiated by the opening of voltage-gated sodium and potassium channels.
- 28 The diagram represents a G-protein coupled receptor on the cell surface membrane. A peptide hormone ligand is bound to the receptor and initiates the production of a second messenger.



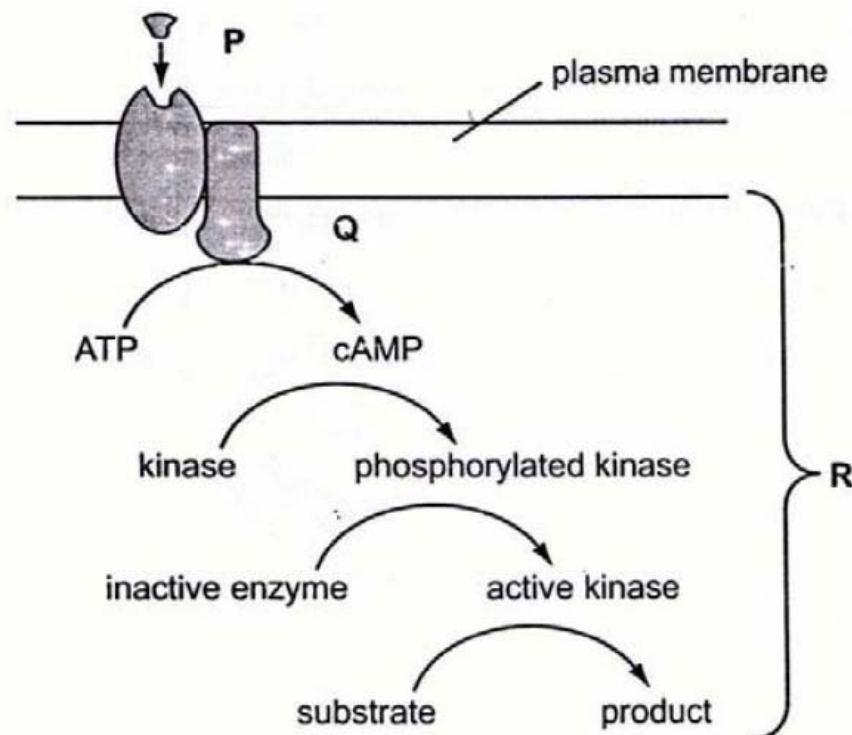
What is the second messenger?

- A a peptide hormone
- B ATP
- C cyclic AMP
- D kinase

- 29 Which of the following is a correct statement regarding the hormonal control of blood glucose level?

	Hormone	Receptor which hormone binds to	Effect
A	Insulin	Tyrosine kinase receptor	Increase glycogenesis
B	Insulin	G protein-linked receptor	Increase gluconeogenesis
C	Glucagon	Tyrosine kinase receptor	Increase gluconeogenesis
D	Glucagon	G protein-linked receptor	Increase glycogenesis

- 30 The diagram below shows an example of cell signaling. Identify the stages **P**, **Q** and **R**.



	P	Q	R
A	amplification	ligand-receptor interaction	phosphorylation
B	ligand-receptor interaction	dephosphorylation	amplification
C	transduction	phosphorylation	amplification
D	ligand-receptor interaction	amplification	dephosphorylation

- 31** Molecular homology of haemoglobin is used to determine evolutionary relatedness in a range of mammals. Either nucleic acids or proteins can be sequenced. The sequence of nucleic acids and proteins are compared.

Which sequence will vary **LEAST** in related mammals?

- A** Amino acid
- B** DNA
- C** Edited mRNA transcript
- D** Primary mRNA transcript

- 32** When organochlorine insecticides such as DDT were in widespread use, mosquitoes in malarial regions developed resistance more rapidly than houseflies in Britain.

What could account for the difference in the rates of the development of resistance?

- A** More insecticides were used in Britain.
- B** More insecticides were used in malarial regions.
- C** Mosquitoes have fewer random mutations when exposed to insecticides.
- D** Mosquitoes have more random mutations when exposed to insecticides.

- 33** Fish in the Arctic and Antarctica have antifreeze proteins. But these two populations of fish diverged long before the antifreeze genes evolved. It was found that the genes that code for the antifreeze proteins in fish, at the Arctic and Antarctica, are quite different. This is evidence that separate and independent episodes of molecular evolution occurred, and obtained the same functional result.

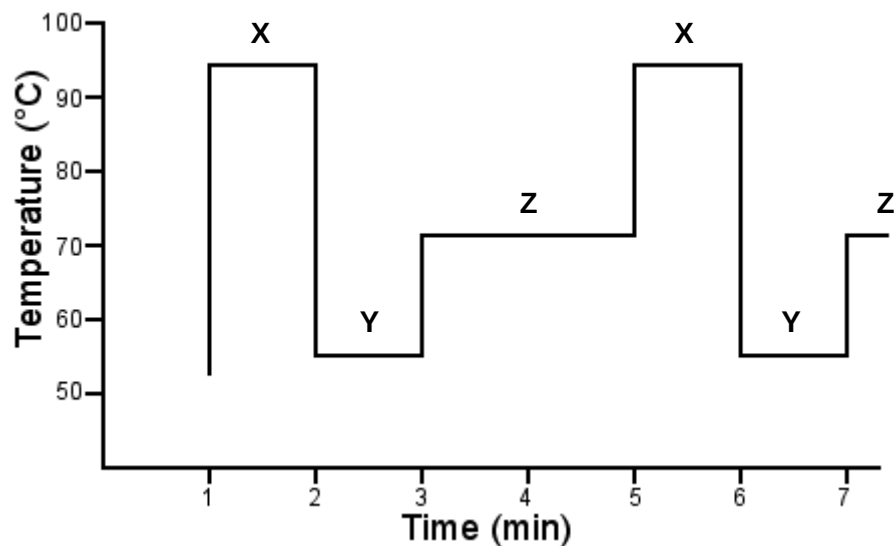
Which of the following processes is responsible for the antifreeze property in the two populations of the fish?

- A** Reproductive isolation
- B** Divergent evolution
- C** Convergent evolution
- D** Geographical isolation

34 Which of the following is **TRUE** of a genomic library?

- A It is a catalogue of the different RFLPs of organisms within a population.
- B It is a collection of DNA fragments that make up the entire genome of a particular organism.
- C It is a collection of DNA fragments created by reverse transcriptase which are then inserted into vectors.
- D It is a catalogue of all genes of an organism's genome which have been sequenced thus far.

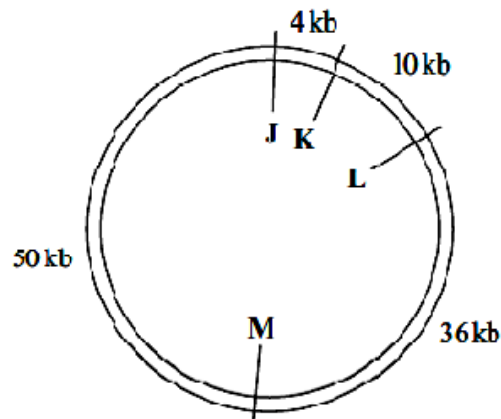
35 The diagram below shows the changes in temperature in a thermocycler over time during polymerase chain reaction.



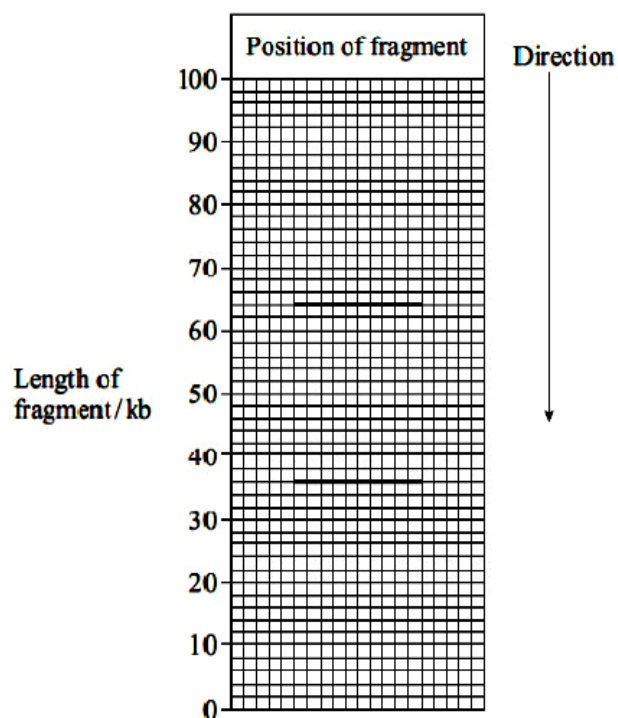
Which of the following statements is **TRUE**?

- A Elongation of new daughter strand occurs during Y.
- B Taq polymerase functions optimally at X.
- C DNA primers are anneal to the DNA template during Z.
- D Double stranded DNA template denatures into single strands during X.

- 36** The diagram below shows the position of four restriction sites J, K, L and M for four different restriction enzymes in a single plasmid. The distances between these sites are measured in kilobases of DNA.



The plasmid was cut using only two restriction enzymes. The resulting fragments were separated by gel electrophoresis. The positions of the fragments are shown in the chart below.

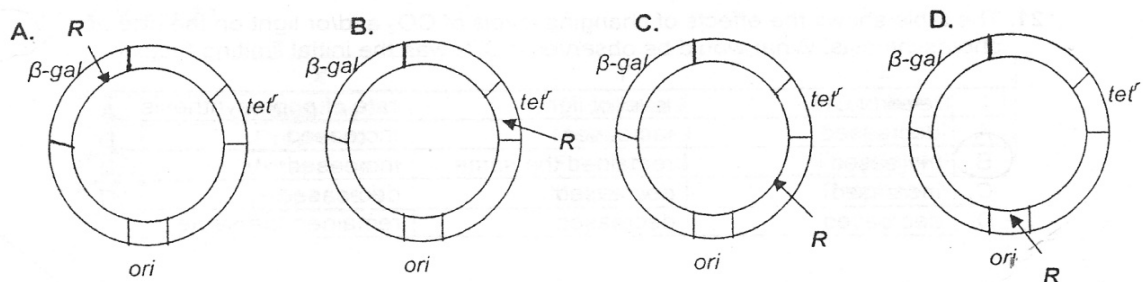


Which restriction sites were cut?

- A** J and K
- B** L and M
- C** J and M
- D** L and K

- 37** A synthetic plasmid having an origin of replication (*ori*), a restriction site (*R*), two genetic markers, β -galactosidase gene (*β -gal*) and tetracycline-resistance gene (*tet^r*), was used as a vector to transfer the human growth hormone gene into bacteria. The *β -gal* gene codes for an enzyme β -galactosidase, which breaks down a colourless compound called X-gal to a blue compound.

At which position should the restriction site (*R*) be located on the plasmid in order to identify the desired transformed bacteria on an agar medium containing both X-gal and tetracycline?



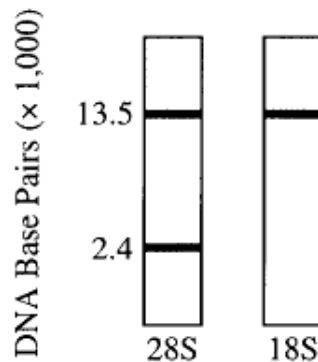
- 38** How is the polymerase chain reaction (PCR) similar to the replication of DNA?

- I DNA is heated to break hydrogen bonds
- II DNA unzips
- III free nucleotides are used
- IV DNA polymerase are required

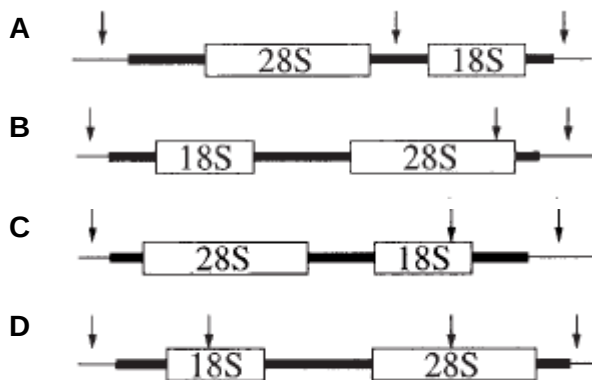
- A I only
- B II and IV only
- C I, II and III only
- D II, III and IV only

- 39 The autoradiograms obtained below (after electrophoresis and Southern Blotting) show human DNA digested with a specific restriction enzyme and probed with labeled rRNA.

On the left, the probe used was 28S rRNA. On the right, the probe used was 18S rRNA.



If the arrows in the following map show the location of the restriction sites of this restriction enzyme, which map **best** explains the results shown above?



- 40 Which of the following explains why insulin expressed in a prokaryotic host cell is not functional?

- I. Prokaryotic cells cannot replicate human DNA.
- II. Prokaryotic cells cannot splice mRNA.
- III. Prokaryotic cells do not possess sER.
- IV. Prokaryotic protein expression machinery stops at translation.

- A I only
- B II only
- C II and III
- D I, II, III and IV

END OF PAPER

2013 Y6 Preliminary Exam H2**MCQ Answer Scheme**

1	C	21	D
2	B	22	B
3	B	23	C
4	D	24	A
5	D	25	C
6	C	26	D
7	A	27	D
8	D	28	C
9	B	29	A
10	B	30	B
11	C	31	A
12	B	32	B
13	C	33	C
14	D	34	B
15	A	35	D
16	A	36	B
17	B	37	A
18	C	38	D
19	B	39	B
20	C	40	B